

Real-Time Driver Drowsiness Detection Using Hybrid CNN-LSTM Model with Facial Feature and Behavioral Analysis

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Abstract—Driver Drowsiness is a leading cause of road accidents. It is responsible for around 20 percent of fatal accidents all around the world. Early detection of driver tiredness is crucial in preventing road accidents and enhancing road safety. There are traditional detection methods like self-reporting, physiological and vehicle-based measures but they are either unreliable or intrusive which limits its effectiveness. In this paper, we have developed a deep learning-based technique to correctly detect driver drowsiness in real time using a non-intrusive method. The proposed approach consists of physiological monitoring and computer vision approach to analyze facial features, head poses, blink duration, blink rate, rate of eye closure and yawning using the camera within the vehicles. These features are processed by deep learning models which are compatible with computer vision. CNN model along with LSTM is used in finding temporal patterns and trained on diverse dataset to predict the drowsiness levels. Results indicate hybrid model CNN-LSTM is effective in detecting drivers drowsiness.

Index Terms—Driver Drowsiness Detection, Eye-Blink Monitoring, Fatigue Detection Systems, Behavioral analysis, Eye Movements, Transfer Learning

I. INTRODUCTION

Driver fatigue is a major contributor to the high frequency of road accidents [1]. Prolonged periods of continuous driving often lead to significant fatigue accumulation. This leads to drowsiness and causes drivers to lose control of vehicle leading to accidents. According to a study, more than 50% (3.5–67.3%) of users agreed that they had been sleep deprived [1].

Developing a non-intrusive device to detect driver drowsiness presents significant challenges. Thousands of people have died due to drowsy drivers, particularly due to drivers of goods vehicles such as trucks. Frequent night trips often result in sleep deprivation, increasing the risk of accidents caused by fatigue.

Intelligence Transportation Systems strive to ensure public safety and reduces road accidents. Major reasons of accidents in rural roads, is due to driver monotony and sleepy behavior during driving. Prolonged driving significantly impairs

a driver's judgment and decision-making abilities. Common causes of driver drowsiness include extended driving hours, alcohol consumption, certain medications and driving after a heavy meal [1].

Many researchers around the globe are trying to find various methods to detect drivers drowsiness and ways to solve the problem. The methods are generally of three types:

1. Physiological Measures

It involves measurements that are obtained by placing intrusive electronic devices onto the skin of drivers. This methods is highly intrusive and include EEG (Electroencephalography), ECG (Electrocardiography) and EOG (Electrooculogram) techniques. High cost and intrusive nature limits its practical applications [2].

2. Vehicle-based Measures

It is based on vehicle control systems, which includes breaking patterns, steering patterns, and lane departure movements. These techniques provides better results and are non-invasive as compared to physiological measures but gives inaccurate results due to dependency on roads and drivers behavior [2].

3. Behavioral Measures

These techniques depend on computer vision measures that is visual- based techniques such as analyzing drivers behavior using visual features like eyes, mouth and head. These techniques are most effective in extracting relevant visual features of drivers that usually defines driver's degree of sleepiness [2], [3].

This study utilized behavioral parameters to detect drowsiness in drivers using deep learning approach. First, we use a parameter based on eyes, namely Eye Aspect Ratio (EAR). It is the ratio of the height of the eye to its width and calculated using facial landmark or shape detection. This parameter is used to measure the openness of eye in each frame and a sharp drop in one frame implies an eye blink. Time-series analysis is

used to identify between a normal or abnormal blink signifies and thus drowsiness [2], [4].

Next, for finding if a person is yawning we have used Mouth Aspect Ratio (MAR). MAR is computed as the ratio of height of mouth usually taken from one end of the lip to another to its width and an abnormal increase in its value for a period implies yawning.

Finally, head tilt or movement detection was utilized to determine if the person feels drowsy using his head positions at different intervals of time. For example, head nodding frequency is directly proportional to drowsy behavior and it increases exponentially due to backward bending. These measures were used to provide a more accurate model which helps in determining whether the driver is drowsy or not in real time.

To address the research gaps identified in previous studies, this study aims to develop a hybrid deep learning model using a CNN-LSTM approach. The model integrates multimodal behavioral parameters, including Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head movements, to enhance the accuracy of driver drowsiness detection.

II. LITERATURE REVIEW

In the past decades, detection systems for driver drowsiness have gained vast attention because of its crucial role in enhancing road safety. Many innovative solutions to real-time driver fatigue detection have been developed under the continuous development of artificial intelligence and deep learning algorithms. The models were proposed with a focus on increasing accuracy, reducing spam alerts, wider availability and integration with modern automobile systems.

Das et al. (2024) [5] presents a cutting-edge method by integrating CNN-LSTM and U-Net architectures for real-time detection of driver drowsiness through facial movement study. Their system achieved an accuracy of 98.8 %, outstripping the performance of other models like VGG-16, GoogLeNet, AlexNet, and ResNet50. The primary advantage of this method is its ability to provide high accuracy while being non-intrusive. But the environmental factors such as lighting conditions and its computationally intensive nature causes the system to be sensitive and experience problems in detection.

Li et al. (2023) [6] studied the method of Support Vector Machine (SVM) for drowsiness detection, accomplishing an accuracy of 91.92%. The integration of SVM with other driver assistance systems is a very practical solution but stumbles at problems like the need for extensive training and the potential of external factors such as stress or fatigue influencing results. Thus they end up demanding a computationally intensive hardware thereby decreasing wider adaptability.

Pauly et al. (2023) [7] also employ Support Vector Machines, along with Histogram of Oriented Gradients (HOG), which can help in detecting drowsiness in real-time with an accuracy of 91.6%. This method's real-time capabilities make it appreciated for early alert systems, though the model's efficacy can be compromised under varying head tilts and lighting conditions. The study is limited by its small datasets

and highlights a research gap of training the classifier on diverse datasets that could improve its robustness in varied environments.

Rahman et al. (2022) [8] studies a real-time eye-blink monitoring system based on Viola-Jones and AdaBoost algorithms, which are enhanced with Haar classifiers, and achieved an accuracy of 94%. This system has problem in its performance being influenced by distraction factor like obstructions and occlusions. The authors emphasize that more advanced monitoring techniques capable of accommodating a broader range of driver behaviors need to be researched and applied for better performance.

Similarly, Coetzer et al. (2022) [9] have developed a study based on Artificial Neural Networks (ANN), SVM, and AdaBoost aiming to develop a hybrid system that detects drowsiness in drivers and achieved an accuracy of 98.1%. Their model performs well under challenging conditions like low lighting, though it faces problems with obstructions and head tilting during eye detection.

Weng et al. (2022) [10] has focused on improving detection accuracy by combining facial expressions with other biometric signals in a Multimodal Emotion Recognition system. This approach has enhanced detection capabilities of the system. But its extensive focus centered on emotion recognition rather than fatigue detection causes it to lack persuasiveness to be developed as a driver drowsiness detection system and further lacks a focus on temporal analysis like blink rate and yawning.

Anjali et al. (2021) [11] has applied the Viola-Jones algorithm on a Haar Cascaded Classifier to develop a reliable method for detecting driver drowsiness. Despite its impressive performance and reliability which can help in reducing drowsiness-related accidents, the extensive training on large datasets has become a bottleneck for it to function effectively in real-time environments. While The model also suffers from lack of robustness.

Mungra et al. (2020) [12] has developed a study that tries to identify whether CNN-based emotion recognition techniques can help in detecting behaviors linked to fatigue. While their method demonstrates high accuracy in recognizing emotional states, the limited exploration of data augmentation techniques and alternative CNN architectures has left a vast room for further research to come up with methods more related to drowsiness detection system as well as to address the gaps of the paper.

Magan et al. (2022) [13] in their study use Dlib, EfficientNet, and Recurrent Neural Networks developing a non-intrusive detection system using cameras. Although this has proven to be effective, the inference from external factors such as obstruction and positional logic causes the system to be prone to extreme observations. One of the suggestion was to try for further development Alternative feature selection techniques to improve detection accuracy.

Lastly, Punitha et al. (2020) [14] states a approach based on Viola-Jones combined with Face Cascade Classifiers and SVM, that through eye-state analysis achieved a 93.5% accuracy in detecting drowsiness. While this method is effective,

ambient features like lighting and occlusions may cause fluctuations in results. It suggests for more diverse dataset to be used and further development to deal with diverse environmental conditions.

The recent developments in driver drowsiness detection systems has revealed significant improvements in accuracy and real-time monitoring. The review of previous studies is discussed in Table 1. Our study aims to tackle the challenges in the current methods. There is a need to focus on the integration of multimodal data and more sophisticated temporal analysis techniques.

III. METHODOLOGY

The proposed methodology combines the configuration of CNN (Convolutional Neural Networks) and LSTM (Long Short-Term Memory) architectures to capture temporal and spatial features from facial landmarks extracted using MediaPipe [15]. The proposed model is shown in Fig. 1. The detailed approach to this method is as follows.

A. Dataset Collection and Preparation

After collection of images from UTARLDD dataset, they were classified and organized into two separate folders, which are ‘drowsy’ and ‘non-drowsy’ states accordingly as their name suggests. The images were preprocessed so that key facial landmarks could be extracted using MediaPipe’s ‘FaceMesh’ module, which captures 468 facial landmarks from these 2D images. For each image, facial landmarks were extracted to compute three key features:

1. Eye Aspect Ratio (EAR): An indicator of blinking behavior, calculated using specific eye landmarks. It captures changes in eye openness [16], [17].
2. Mouth Aspect Ratio (MAR): Captures mouth movements, specifically yawning, by analyzing distances between mouth landmarks [18].
3. Head Pose Ratio: Analyzes the relative positions of the nose and chin, indicating head tilts and movement patterns relevant to drowsiness [19].

B. Landmark Extraction

Facial landmarks were extracted using MediaPipe, and EAR, MAR, and head pose were calculated using custom functions to track key facial features associated with drowsiness. For each image, a feature vector was formed by concatenating the facial landmarks and the three features (EAR, MAR, head pose).

C. Feature Scaling

The StandardScaler has been used to normalize the features, with the aim of achieving a mean of 0 and a standard deviation 1. This process is crucial to enhance the model’s performance.

TABLE I
REVIEW OF PREVIOUS STUDIES ON DRIVERS DROWSINESS DETECTION

Ref.	Methodology	Accuracy	Advantages	Limitations/ challenges
[5]	CNN-LSTM	98.8%	High accuracy, real-time detection, non-intrusive, integrates with ADAS systems	Limited exploration of different CNN architectures and temporal dynamics integration
[6]	SVM	91.92%	Can integrate with other driver assistance systems	Need for improved generalization in real-world driving scenarios
[7]	Histogram of Oriented Gradient, SVM	91.6%	Detects drowsiness in real-time	Difficulty in detecting under different head poses and lighting conditions
[8]	Viola-Jones, AdaBoost, Haar Classifier	94%	Real-time detection through eye-blink monitoring	More sophisticated monitoring needed for different driver behaviours
[9]	Artificial Neural Networks, SVM, AdaBoost	98.1%	Challenging environmental conditions like poor lighting can be handled	Lack of robustness in real-world driving situations
[10]	Multimodal Emotion Recognition	RR - 93%	Improved accuracy through fusion of multiple modalities	Improper use of deep learning architectures and lack of focus on drowsiness detection through temporal analysis
[11]	Viola-Jones, Haar Cascaded Classifier	-	Can minimize accidents caused by drowsiness	Needs improvement in robustness for general use in real-time systems
[12]	CNN-based Emotion Recognition	78.52	High accuracy in detecting emotional states	Investigation needed on the impact of different CNN architectures
[13]	CNN	93.0%	Non-intrusive detection using cameras	Limited exploration of alternative feature extraction techniques
[14]	Viola-Jones, Haar Cascade Classifiers, SVM	93.5%	Effective analysis on the status of eyes	Integration with other data sources for more accurate detection

D. Data Splitting

The whole dataset was split into 3 sets training, validation, and test sets using a stratified approach to ensure balanced representation of both ‘drowsy’ and ‘non-drowsy’ across each split. This was used to further tune in the model using these sets, whereas the test set was reserved for final evaluation. [19]

E. Model Architecture

CNN and LSTM were combined to create a hybrid architecture that took into account the data's temporal and spatial dependencies.

- **CNN Layers:** The CNN component extracts spatial features from the facial landmarks. To find spacial patterns in the input, two 1 - Dimensional convolution layer (1D-convolution) were employed, followed by max-pooling layer.
- **Fully Connected Layers:** To learn high-level feature representation, the output of the CNN model is flattened and pass to the dense layer.
- **LSTM Layers:** The CNN model's output was reshaped and pass to a bidirectional LSTM (Long Short Term Memory) layer to capture temporal dependencies in the extracted feature across time, such as prolonged tilt head position or eye closure or yawning.
- **Output Layer:** A final dense layer with a sigmoid activation function was used for binary classification ('drowsy' vs. 'non-drowsy').

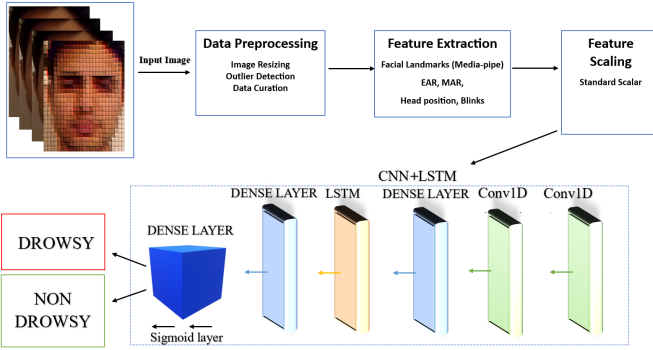


Fig. 1. Model Architecture

F. Training

Adam optimizer and binary cross-entropy loss function were used to train the model, which prevent over fitting and generalize the model, Early stopping and learning rate reduction were employed. Early stopping kept track of the validation loss and restored the best weights till the step after 15 epochs without improvement, as a result of a 0.2 reduction in the learning rate following 5 epochs of plateauing.

G. Evaluation Metrics

- **Accuracy:** Accuracy was tracked throughout the training process for both the training and validation sets.
- **Confusion Matrix:** Generated to visualize the performance of the model on the test set.
- **ROC and AUC:** To examine the trade-off between drowsiness detection and non-drowsiness detection, the ROC curve was plotted. In order to facilitate comparison and provide a scalar measure of the model's classification performance, the area under the curve was also computed.

- **Classification Report:** Calculation of Recall, precision, and F1 scores allows to easily assess the model's performance in contrasting and classifying between the two classes, which are drowsy and non-drowsy.

IV. RESULTS AND DISCUSSIONS

This section discussed the results obtained after dividing the dataset into training and validation sets for analysis and applied Deep Learning technique (CNN-LSTM) to predict driver drowsiness.

A. Confusion Matrix

- **Confusion Matrix:** A heatmap was used to display the confusion matrix (Fig. 2), summarizing the number of correct and incorrect predictions making visualization of model's performance easier.

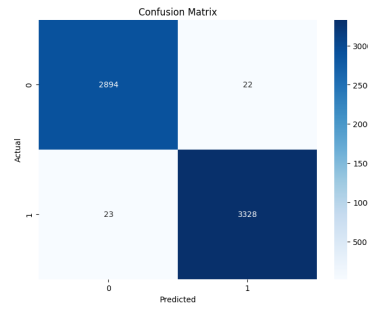


Fig. 2. Confusion matrix

B. Training and Validation Curves

- **Training and Validation Curves:** Plots of training and validation accuracy and loss were generated to illustrate the model's learning progress and shown in Fig. 3.

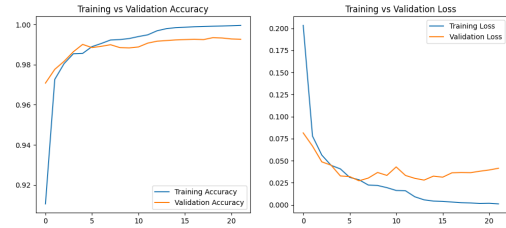


Fig. 3. Training and Validation Curves

C. ROC Curve

- **ROC Curve:** The curve was plotted in Fig. 4 to display the performance trade-offs, along with the AUC (Area Under the Curve) score for overall performance evaluation. This helps in detecting at which threshold we can find the better rate of system finding drowsiness with some trade-offs for the detection of non-drowsiness from our model and thus finding appropriate results.

The final result of the model are represented in Fig. 5 and Fig. 6 representing the drowsy and non-drowsy states respectively.

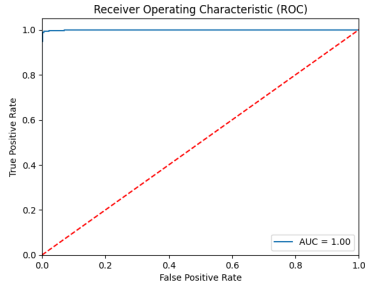


Fig. 4. ROC curve

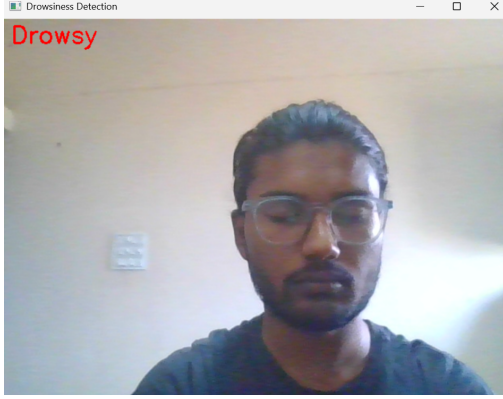


Fig. 5. Drowsy State Detection

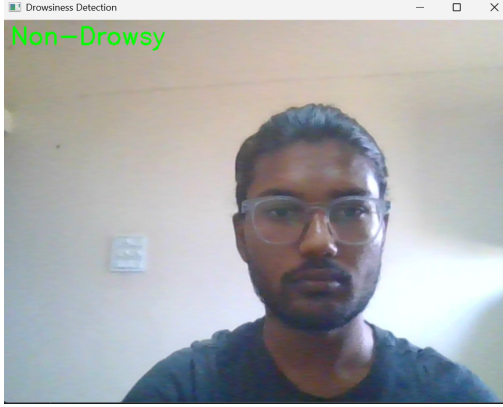


Fig. 6. Non - Drowsy State Detection

The CNN model with the simplest configuration and the hybrid model (CNN+LSTM), achieves the lowest accuracy. This result suggests that when CNN is used solely for feature extraction, the model treats the entire image as a feature set, which reduces its effectiveness and accuracy.

By integrating the MediaPipe framework for feature extraction, the hybrid model (MediaPipe+CNN+LSTM) demonstrates a significant improvement, with the accuracy score increasing substantially. This enhancement suggests that incorporating MediaPipe-based feature extraction improves the model's performance by focusing on crucial landmarks and positional information. However, the model still suffers from

a lack of dedicated features to provide better input for the CNN-LSTM architecture.

So, after addition of dedicated features the model achieved high recall and precision. The most comprehensive hybrid model(MediaPipe + EAR + MAR + Head Position + CNN +LSTM) achieved the highest accuracy, with an accuracy score of 98.89% and an F1 score of 98.00%.

This configuration points up the benefits of combining MediaPipe feature extraction with additional metrics-MAR, EAR and head position which when fed to CNN and LSTM provides an astounding increase in accuracy and precision. Each included feature extraction component enriches the feature set, allowing the model to capture subtle and complex patterns in data, which results in superior accuracy and overall performance. The comparison of results obtained from proposed work with similar studies is presented in Table 2.

TABLE II
COMPARISON OF PROPOSED WORK WITH PREVIOUS STUDIES

Ref.	Methodology	Accuracy
[5]	CNN-LSTM	98.80%
[13]	CNN	93.0%
[18]	CNN	96.32%
This study	CNN-LSTM-Mediapipe- EAR-MAR- Head pose	98.89%

V. CONCLUSION AND FUTURE WORK

In this study, a real-time driver drowsiness detection system was proposed using a deep learning method that combines CNN and LSTM models. Through the use of MediaPipe's facial landmark detection we extracted features like EAR, MAR and head tilt. This allowed our system to effectively identify the signs of driver drowsiness. By analyzing temporal patterns in blinking, yawning and head positioning the proposed model helps achieve a remarkably high accuracy in detecting drowsiness. This hybrid architecture ensures that the detection process is reliable and adaptable to diverse positions and varying head poses, providing a feasible solution for enhancing road safety. While our system demonstrates significant accuracy, it faces challenges, particularly with environmental factors such as poor lighting or obstructions, which can affect detection and degrade performance. Additionally, although real-time processing is feasible, it is slow and requires high computational resources. Therefore, further optimizations are necessary to make it practical for widespread use in all vehicles.

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